

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

**ENVIRONMENTAL SCIENCE****Paper 1 (Generated practice paper - Set 37)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

**Time allowed: 3 hours****Materials**

For this paper you may use:

- a calculator.

**Instructions**

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

**Answer all questions in the spaces provided.**

|   |   |
|---|---|
| 0 | 1 |
|---|---|

A rural community in Pakistan is considering a low-head helical (Archimedes screw) turbine on a river rather than joining a high-head dam scheme.

|   |   |
|---|---|
| 0 | 1 |
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**1**

Explain why low-head turbines are more suitable than high-head HEP schemes for many rural areas.

**[5 marks]**

|   |   |
|---|---|
| 0 | 1 |
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2

Suggest **one** reason why such schemes may not be installed even at suitable sites.

**[1 mark]**

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| 0 | 1 |
|---|---|

3

The maximum flow rate at the site is  $6.5 \text{ m}^3 \text{ s}^{-1}$  giving an output of 100 kW at 90% turbine efficiency. At 10% of maximum flow the efficiency falls to 40%. Calculate the output at 10% of maximum flow.

**[2 marks]**

### Answer

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kW

0 1 . 4

Outline **two** ways a high-head reservoir can affect the river downstream.

[2 marks]

1 .....

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2 .....

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0 2 . 1

Complete Table 1 to show the source and disposal method for each category of radioactive waste from nuclear electricity generation.

[6 marks]

Table 1

| Waste category           | Example / source | Storage or disposal method |  |
|--------------------------|------------------|----------------------------|--|
| High-level waste         |                  |                            |  |
| Intermediate-level waste |                  |                            |  |
| Low-level solid waste    |                  |                            |  |

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0 2 . 2

Explain why high-level waste must be cooled and why its storage must be monitored for very long periods.

[2 marks]

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0 2 . 3

Describe how low-level liquid and gaseous wastes are treated before discharge.

[2 marks]

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0 3 . 1

Explain why it is difficult to be certain that a particular flood or heatwave is caused by anthropogenic climate change.

[3 marks]

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0 3 . 2

Explain why historic temperature records collected in towns may overestimate global warming.

[2 marks]

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03 . 3

Describe how each of the following sources of proxy data is used to estimate past climates and give one limitation.

[5 marks]

**Tree rings (dendrochronology)** .....

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**Ice cores** .....

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04

The Aral Sea in central Asia was once the fourth largest lake in the world. From the 1960s water from the two rivers feeding it (the Amu Darya and Syr Darya) was diverted to irrigate cotton and rice. Table 1 shows changes in the lake.

**Table 1**

| Year                                    | 1960   | 1980   | 2000   | 2015     |
|-----------------------------------------|--------|--------|--------|----------|
| Lake area / km <sup>2</sup>             | 68 000 | 52 000 | 24 000 | 7 000    |
| Salinity / g dm <sup>-3</sup>           | 10     | 18     | 45     | over 100 |
| Commercial fish catch / thousand tonnes | 40     | 10     | 0      | 0        |

04 . 1

Calculate the percentage reduction in the area of the Aral Sea between 1960 and 2015.

[2 marks]

Answer

%

0 4 . 2

Explain why the salinity of the lake increased as the area fell.

[2 marks]

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0 4 . 3

Explain why the fish catch fell to zero, and describe **two** other environmental consequences of the shrinking of the lake.

[4 marks]

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0 4 . 4

Suggest why disagreements over the rivers continue between the countries of the region.

[2 marks]

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**10****0 5 . 1**

Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

**[1 mark]**

A Absorbent mops

☐

B Steam washing

☐

C Inflatable booms

☐

D Polymerising agents

☐**0 5 . 2**

Explain how polymerising agents make an oil spill easier to clean up.

**[2 marks]**

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**0 5 . 3**

Explain why deoxygenation is a more common problem in water than in air.

**[2 marks]**

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**5**

**0 6**

Figure 1 shows a simplified nitrogen cycle with some processes labelled 1 to 6.

**Figure 1**

| Number | Process description                                                            |
|--------|--------------------------------------------------------------------------------|
| 1      | Nitrogen gas converted to ammonium ions by Rhizobium in root nodules           |
| 2      | Ammonium ions oxidised to nitrite then nitrate by Nitrosomonas and Nitrobacter |
| 3      | Nitrates reduced to nitrogen gas by Pseudomonas in anaerobic soil              |
| 4      | Amino groups in dead organic matter released as ammonium ions by decomposers   |
| 5      | Nitrogen and oxygen react during lightning to form NO <sub>x</sub>             |
| 6      | Nitrates washed out of soil into rivers                                        |

**0 6****1**

Name processes 1 to 6.

**[3 marks]**

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**0 6****2**

Explain why draining a waterlogged field increases the nitrate available to crops.

**[3 marks]**

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**0 6 . 3**

Explain why nitrogen is described as a resource that is 'not very abundant' for living organisms even though the atmosphere is 78% nitrogen.

**[2 marks]****0 6 . 4**

Explain why growing legumes such as clover in a crop rotation reduces the need for artificial fertiliser.

**[2 marks]****10****0 7**

Table 1 shows typical noise levels of some sounds.

**Table 1**

| Sound                                   | Noise level / dB |
|-----------------------------------------|------------------|
| Threshold of human hearing (at 1000 Hz) | 0                |
| Leaves rustling                         | 20               |
| Library                                 | 40               |
| Normal conversation                     | 60               |
| Busy road                               | 80               |
| Chain saw                               | 110              |
| Jet engine at 30 m                      | 140              |

**0 7 . 1**

Explain why a logarithmic scale is used to measure sound.

**[2 marks]****0 7 . 2**

Use Table 1 to calculate how many times greater the sound intensity of a chain saw is than that of a busy road.

**[2 marks]**

Answer  times

07 . 3

A machine produces 85 dB. A second identical machine is switched on next to it. State the new noise level and explain your answer.

[2 marks]

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07 . 4

Explain what is meant by the statement that 0 dB does not mean there is no sound.

[1 mark]

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07 . 5

State the frequency range of hearing for a typical young person and explain why the dB(A) scale is used for human hearing.

[3 marks]

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**0 8**

Table 1 shows the annual emissions of two pollutants in the UK in 1970 and 2015.

**Table 1**

| <b>Pollutant</b> | <b>Emissions in 1970<br/>/ thousand tonnes</b> | <b>Emissions in 2015<br/>/ thousand tonnes</b> | <b>Main source in<br/>2015</b> |
|------------------|------------------------------------------------|------------------------------------------------|--------------------------------|
| Sulfur dioxide   | 6 400                                          | 220                                            | Energy industries              |
| Nitrogen oxides  | 2 700                                          | 930                                            | Road transport                 |

**0 8****1**

Calculate the percentage reduction in sulfur dioxide emissions between 1970 and 2015.

**[2 marks]**

Answer

%

**0 8****2**Suggest **three** reasons why sulfur dioxide emissions fell by a much larger percentage than nitrogen oxide emissions.**[3 marks]****1**

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**2**

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**3**

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**0 8****3**Explain why a reduction in NO<sub>x</sub> emissions would also reduce concentrations of tropospheric ozone and PANs.**[2 marks]**

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**0 8****4**

Explain why lichens are used to monitor sulfur dioxide pollution.

**[3 marks]**

10

- 09 Table 1 shows the mean dry mass of wheat plants grown with two fertiliser treatments (10 replicates each).

Table 1

| Treatment                | Mean dry mass / g | Standard deviation / g |
|--------------------------|-------------------|------------------------|
| Organic manure           | 14.2              | 3.9                    |
| Inorganic NPK fertiliser | 16.1              | 4.1                    |

- 09 . 1 Explain what the standard deviation values tell you about the data.

[2 marks]

- 09 . 2 Name the statistical test that should be used to decide whether the difference between the two means is significant, and justify your choice.

[2 marks]

- 09 . 3 The test gave a probability  $p = 0.30$ . State what this means and what the student should conclude.

[2 marks]

09 . 4

Explain why a statistically significant correlation does not prove a causal relationship.

[2 marks]

09 . 5

Suggest how the student could increase the chance of detecting a real difference if one exists.

[2 marks]

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10 . 1

Discuss how an understanding of the properties of pollutants can be used to predict their behaviour in the environment and to plan strategies to control them.

[9 marks]

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Extra space

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**10** . **2** State what is meant by **bioaccumulation**.

**[1 mark]**

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**10**

**Write an essay on one of the following topics.**

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

**[25 marks]**

**OR**

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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Handwriting practice area with 25 horizontal dotted lines.

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**END OF QUESTIONS**